Run report: XL vs MutantXL closure on committed descent draws

Study `research/dreg_closure_mutantxl_20261016/` (pre-registered in `PREREGISTRATION.md`, commit `4966c1190`). **Run** 2026-10-16T02:35Z to 02:39Z UTC (scored run); soundness control 2026-10-16 about 02:42 to 03:21 UTC. **Class** stage diagnostic: no S, no rho ratio, no speed claim.

What ran

- Command: `research/dreg_closure_mutantxl_20261016/run.sh BINARY`, which runs `examples/dreg_closure --max-cols 70000` once per input file. The input is the 41 committed run files of studies 2026-09-23 to 2026-10-06. Binary SHA-256 `f1734a17…a0011` (`runs/binary.sha256`).
- Systems: the chained S_3 decomposition systems (b = 1) of the E_0 descent studies. For m = 3 there are 14 cells (N = 11–25); for m = 4 there are 9 cells (N = 12–32). There are four unsatisfiable draws per cell, 92 in total, each rebuilt from its committed subspace basis and $x(R)$.
- Arms, at each degree D from 3 to the committed XL degree, with $C(N, \leq D) \leq 70000$:
  - plain XL;
  - the MutantXL fall closure (`ReducedBasis::from_system_closed_full`), saturated.
- Host: the 4-core, 16 GB container, which is not isolated (`AGENTS.md` §10). Wall times are practicality notes only.

Completion

- 41 of 41 files exit 0, and no file hit the 3 h limit (`runs/queue.log`).
- 92 of 92 draws were measured.
- The XL-identity assertion held on every degree built: the plain basis refutes exactly at the committed degree.
- Every closure saturated.
- Peak memory 1.0 GB; slowest single degree 39 s.

Measured values

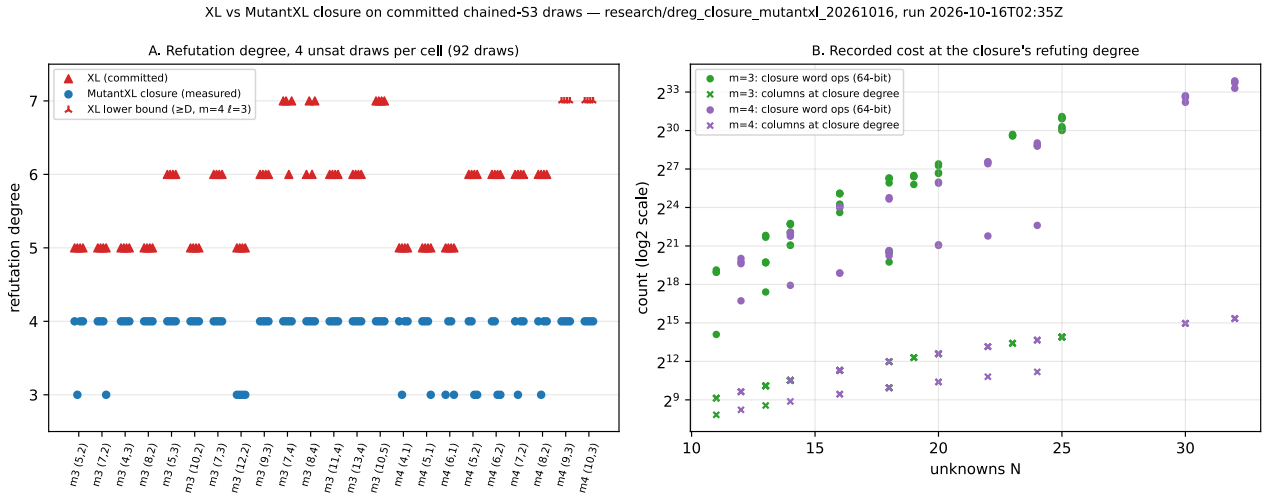


Figure 1: Recorded values from `runs/*.jsonl` (92 draws, run 2026-10-16T02:35Z). A: committed XL refutation degree (triangles; the tri-star marker is a $\geq$ bound) against the measured closure refutation degree (dots), four draws per cell. B: the recorded columns and counted 64-bit word ops of the closure build at its refuting degree, against the unknowns N. Source: `plot.py` (layout only).

- The closure refutation degree is 3 or 4 on every draw.
- The committed XL degree is 5, 6, 7 or ≥ 7.
- Per-cell values, gaps and the registered verdicts are in `score-output.txt` (native `dreg_score closure`):
  - Q1 below in 20 of 20 testable cells, and in the 3 exploratory cells;
  - Q2: 26 draws with gap 1 and 58 with gap ≥ 2;
  - Q3: m = 3 medians 4, 4, 4, 4 over ℓ = 2–5.

Control

Soundness, added after the scored run and disclosed as such: every committed satisfiable draw in the same files, at degrees 3–5 within the cap. 0 of 345 satisfiable draws refuted by XL or the closure, all built to degree 5 (one above every closure refutation). Run in two invocations: the first was killed by the session tool's background time limit after 315

complete draws; the remaining 30 ran to exit 0 (`satisfiable-control-2`, peak 12.3 GB). The kill is a tool limit, not evidence. Records: `runs/satisfiable-control{,-2}.{jsonl,log}`; binary SHA-256 `dcffc6b6…cc094`.

Raw records

`runs/*.jsonl` (one line per draw, with every degree's XL and closure ranks, columns, products, word ops and seconds), `runs/*.log`, `runs/queue.log`. The interpretation is in `../../RESULTS.md`.